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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Develop a Power BI dashboard showing sales, revenue, profit and product performance.

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Experiment No.: 3

Developing a Power BI Dashboard showing sales, revenue, profit and product performance.

GitHub Repository Link: https://github.com/nidhidev2122/Business_analytics

Problem Statement

A retail company stores its business data across flat files (Excel/CSV) and a MySQL database. As a data analyst, connect Power BI Desktop to these data sources, import the required data, explore the Power BI interface, and create basic visuals to analyze sales by product, store, customer, and time.

Methodology / Steps Followed

Step 1: Data Collection

The AdventureWorks dataset was used for this experiment. The dataset consists of multiple files covering sales transactions, customer information, product details, territory data, and calendar information.

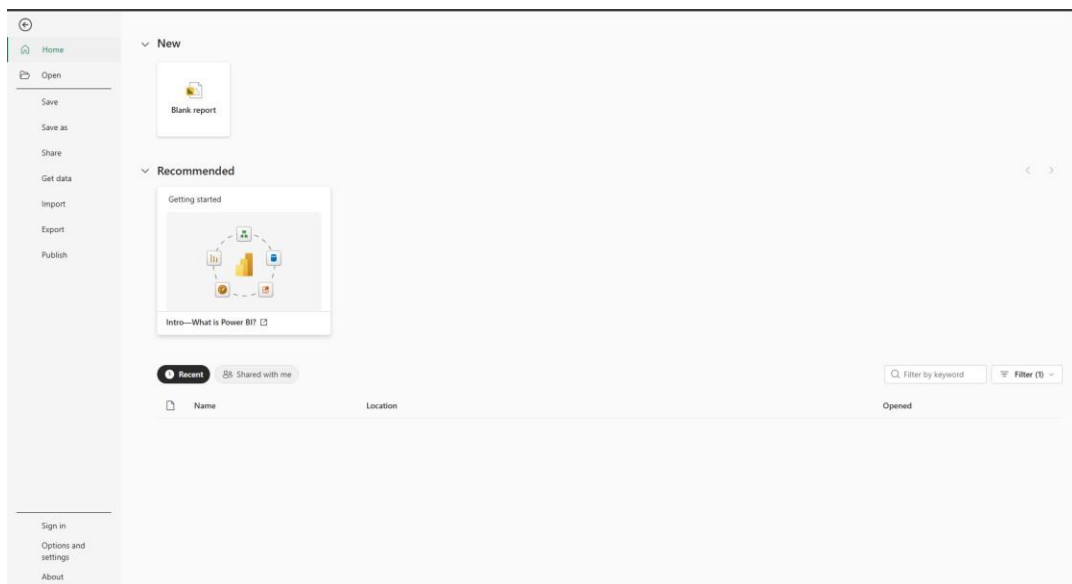


Figure 1: Power BI Desktop Home Screen

Step 2: MySQL Database Setup

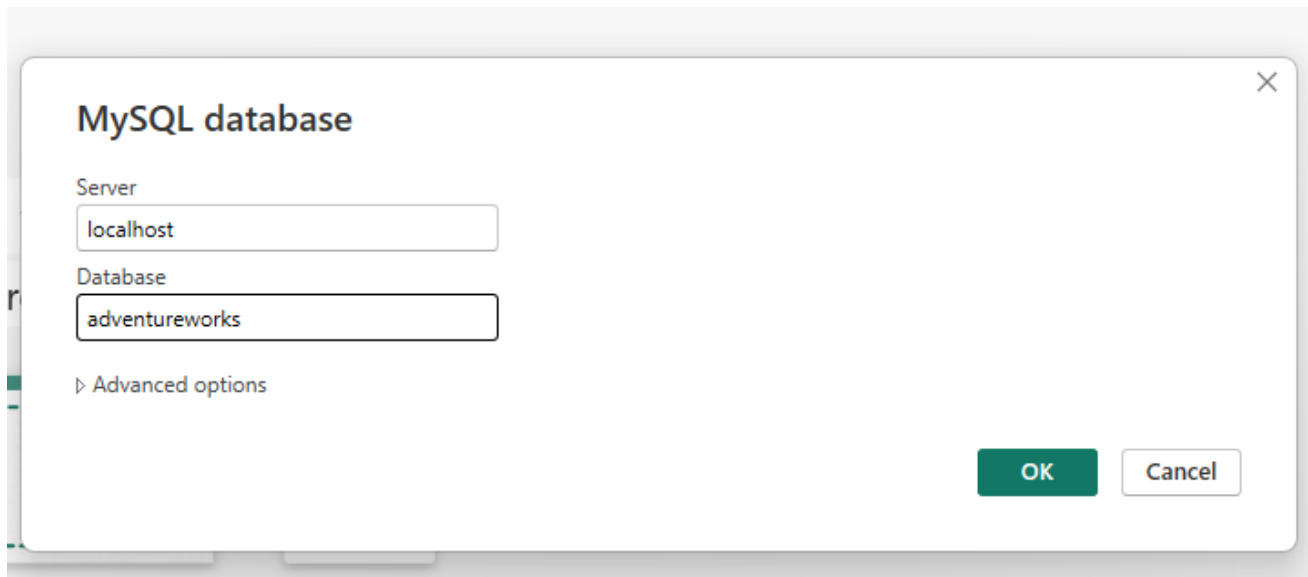


Figure 2: MySQL Database Connection in Power BI

Step 3: Data Import in Power BI

Power BI Desktop was opened and data was imported from two types of sources using the Get Data option:

- **MySQL Database** - Server: localhost, Database: adventureworks. The Sales table containing 56,046 records was loaded into Power BI.
- **CSV Flat Files** - Customer Lookup, Product Lookup, Territory Lookup, and Calendar Lookup files were imported directly using the Text/CSV connector.

The data types of OrderDate and StockDate columns in the Sales table were converted from Text to Date format in Power Query Editor.

Queries [9] fx = Table.TransformColumnTypes(adventureworks_sales,{"OrderDate", type date}, {"StockDate", type date})

	OrderDate	StockDate	OrderNumber	ProductKey	CustomerKey	TerritoryKey	OrderLineItem	OrderQuantity
1	01-01-2022	13-12-2021	S061285		529	23791	1	2
2	01-01-2022	24-09-2021	S061285		214	23791	1	1
3	01-01-2022	04-09-2021	S061285		540	23791	1	1
4	01-01-2022	28-09-2021	S061301		529	16747	1	2
5	01-01-2022	21-10-2021	S061301		377	16747	1	1
6	01-01-2022	23-10-2021	S061301		540	16747	1	1
7	01-01-2022	04-09-2021	S061269		215	11792	4	1
8	01-01-2022	21-10-2021	S061269		229	11792	4	1
9	01-01-2022	24-10-2021	S061286		528	11530	6	2
10	01-01-2022	27-09-2021	S061286		536	11530	6	2
11	01-01-2022	23-10-2021	S061298		530	18155	10	2
12	01-01-2022	02-12-2021	S061298		214	18155	10	1
13	01-01-2022	15-12-2021	S061298		223	18155	10	1
14	01-01-2022	01-10-2021	S061310		538	13541	8	2
15	01-01-2022	08-11-2021	S061310		584	13541	8	1
16	01-01-2022	06-12-2021	S061270		485	18259	9	2
17	01-01-2022	31-10-2021	S061289		528	23694	1	2
18	01-01-2022	16-11-2021	S061289		536	23694	1	2
19	01-01-2022	20-11-2021	S061289		215	23694	1	1
20	01-01-2022	12-10-2021	S061278		477	22308	10	2
21	01-01-2022	28-09-2021	S061278		479	22308	10	1
22	01-01-2022	10-12-2021	S061278		488	22308	10	1
23	01-01-2022	25-09-2021	S061278		580	22308	10	1
24	01-01-2022	01-10-2021	S061305		583	20236	9	1
25	01-01-2022	13-09-2021	S061283		529	26273	1	2
26	01-01-2022	27-09-2021	S061283		538	26273	1	2
27	01-01-2022	03-09-2021	S061283		220	26273	1	1
28	01-01-2022	23-11-2021	S061288		528	19709	6	2
29	01-01-2022	30-10-2021	S061288		536	19709	6	2
30	01-01-2022	21-11-2021	S061288		220	19709	6	1
31	01-01-2022	05-11-2021	S061272		480	21239	9	2
32	01-01-2022	08-09-2021	S061272		528	21239	9	2

COLUMNS: 999+ ROWS: Column profiling based on top 1000 rows

Figure 3: Sales Data Table

Queries [9] fx = Table.TransformColumnTypes(#Promoted Headers,{"CustomerKey", Int64.Type}, {"Prefix", type text}, {"FirstName", type text}, {"LastName", type text}, {"BirthDate", type date}, {"MaritalStatus", type text}, {"Gender", type text}, {"Email", type text}, {"AnnualIncome", type number}, {"TotalChildren", type number}, {"Education", type text})

	CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus	Gender	Email	AnnualIncome	TotalChildren	Education
1	11001	MR.	JON	YANG	08-04-1966	M	M	jon24@adventure-works.com	90000	2	Bachelors
2	11001	MRS.	EUGENE	HUANG	14-05-1965	S	M	eugene10@adventure-works.com	60000	3	Bachelors
3	11002	MRS.	RUBEN	TORRES	12-08-1965	M	M	ruben35@adventure-works.com	60000	3	Bachelors
4	11003	MS.	CHRISTY	ZHU	15-03-1969	S	F	christy12@adventure-works.com	70000	0	Bachelors
5	11004	MRS.	ELIZABETH	JOHNSON	08-08-1968	S	F	elizabeth5@adventure-works.com	80000	5	Bachelors
6	11005	MR.	JULIO	RUIZ	05-08-1965	S	M	julio1@adventure-works.com	70000	0	Bachelors
7	11007	MR.	MARCO	MENTA	09-05-1964	M	M	marco14@adventure-works.com	60000	3	Bachelors
8	11008	MRS.	ROBIN	VERHOFF	07-07-1964	S	F	rob4@adventure-works.com	60000	4	Bachelors
9	11009	MR.	SHANNON	CARLSON	01-04-1964	S	M	shannon38@adventure-works.com	70000	0	Bachelors
10	11010	MS.	JACQUELYN	SUAREZ	06-02-1964	S	F	jacquelyn20@adventure-works.com	70000	0	Bachelors
11	11011	MR.	CURTIS	LU	04-11-1963	M	M	curtis9@adventure-works.com	60000	4	Bachelors
12	11012	MRS.	LAUREN	WALKER	18-01-1968	M	F	lauren41@adventure-works.com	100000	2	Bachelors
13	11013	MR.	IAN	JENKINS	06-08-1968	M	M	ian47@adventure-works.com	100000	2	Bachelors
14	11014	MRS.	SYDNEY	BENNETT	09-05-1968	S	F	sydney23@adventure-works.com	100000	3	Bachelors
15	11015	MS.	CHLOE	YOUNG	27-02-1979	S	F	chloe23@adventure-works.com	30000	0	Partial Coll
16	11016	MR.	WYATT	HILL	28-04-1979	M	M	wyatt32@adventure-works.com	30000	0	Partial Coll
17	11017	MRS.	SHANNON	WANG	26-06-1944	S	F	shannon31@adventure-works.com	20000	4	High Schoc
18	11018	MR.	CLARENCE	RAI	09-10-1944	S	M	clarence22@adventure-works.com	30000	2	Partial Coll
19	11019	MR.	LUKE	LAL	07-03-1978	S	M	luke18@adventure-works.com	40000	0	High Schoc
20	11020	MR.	JORDAN	KING	20-09-1978	S	M	jordan75@adventure-works.com	40000	0	High Schoc
21	11021	MS.	DESTINY	WILSON	03-09-1978	S	F	destiny7@adventure-works.com	40000	0	Partial Coll
22	11022	MR.	ETHAN	ZHANG	12-10-1978	M	M	ethan20@adventure-works.com	40000	0	Partial Coll
23	11023	MR.	SETH	EDWARDS	11-10-1978	M	M	seth46@adventure-works.com	40000	0	Partial Coll
24	11024	MR.	RUSSELL	KIE	17-09-1978	M	M	russell7@adventure-works.com	60000	0	Partial Coll
25	11025	MRS.	ALEJANDRO	BECK	23-12-1945	M	NA	alejandro45@adventure-works.com	20000	2	Partial Coll
26	11026	MR.	HAROLD	SAI	03-04-1946	S	M	harold3@adventure-works.com	30000	2	Partial Coll
27	11027	MR.	JESSIE	ZHAO	07-12-1946	M	M	jessie16@adventure-works.com	30000	2	Partial Coll
28	11028	MRS.	JILL	JIMENEZ	11-04-1946	M	F	jill13@adventure-works.com	30000	2	Partial Coll
29	11029	MR.	JIMMY	MORENO	21-12-1946	M	M	jimmy9@adventure-works.com	30000	2	Partial Coll
30	11030	MRS.	BETHANY	YUAN	22-02-1947	M	F	bethany10@adventure-works.com	20000	2	Partial Coll
31	11031	MRS.	THERESA	RAMOS	22-08-1947	M	F	theresa13@adventure-works.com	20000	4	High Schoc
32											

Figure 4: Customer Lookup Table

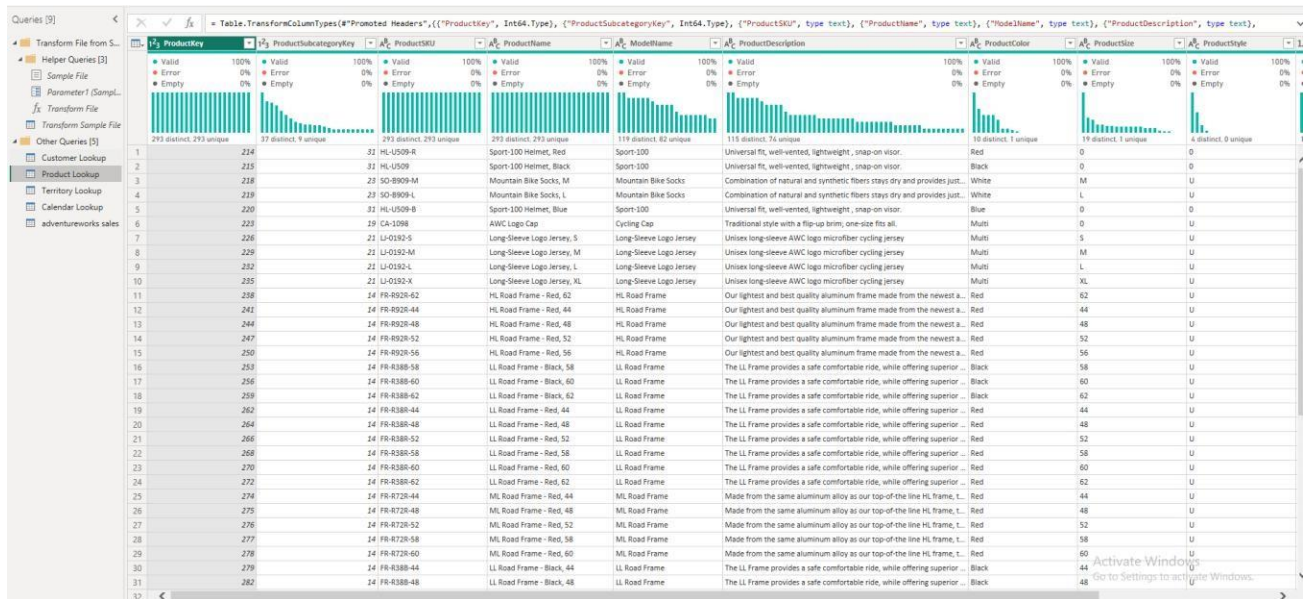


Figure 5: Product Lookup Table

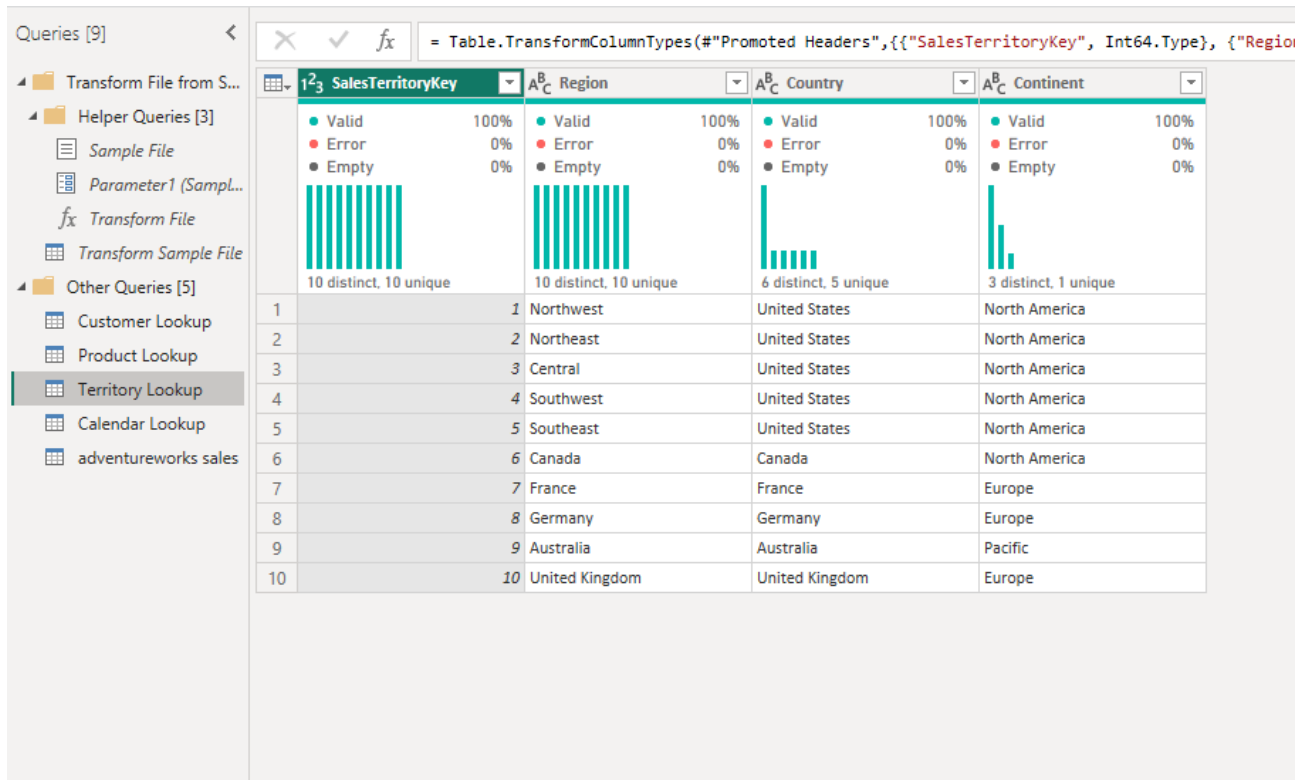


Figure 6: Territory Lookup Table

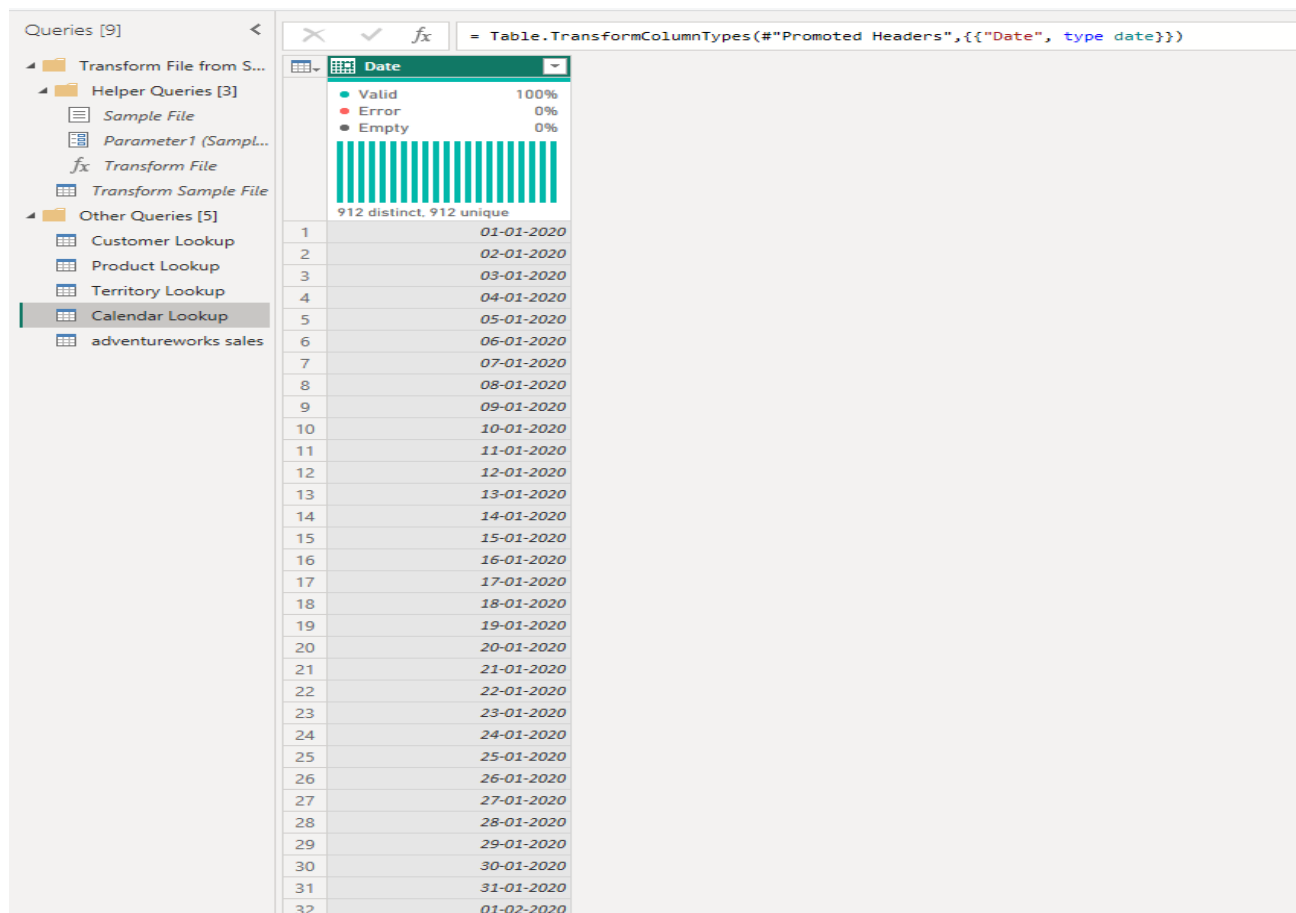


Figure 7: Calendar Lookup Table

Step 4: Data Modeling

After importing all tables, relationships were established between the fact table and lookup tables in the Model View to create a Star Schema:

- Sales → Customer Lookup (via CustomerKey)
- Sales → Product Lookup (via ProductKey)
- Sales → Territory Lookup (via TerritoryKey)
- Sales → Calendar Lookup (via OrderDate → Date)

This star schema ensures efficient filtering and accurate aggregation across all visuals in the dashboard.



Figure 8: Data Model — Star Schema in Model View

Step 5: Dashboard Development

An interactive dashboard titled "AdventureWorks Sales Performance Dashboard" was developed in the Report View. The following visuals were created:

- **3 KPI Cards** - Total Orders (25.164K), Total Quantity Sold (84K), Total Customers (18.149K)
- **Sales by Product** - Clustered Bar Chart showing quantity sold per product
- **Sales by Customer Occupation** - Donut Chart showing sales distribution across customer segments
- **Sales Trend Over Time** - Line Chart showing yearly sales growth from 2020 to 2022
- **Sales by Store Location** - Filled Map showing geographical distribution of sales
- **2 Slicers** - Region and Year for interactive filtering

Output

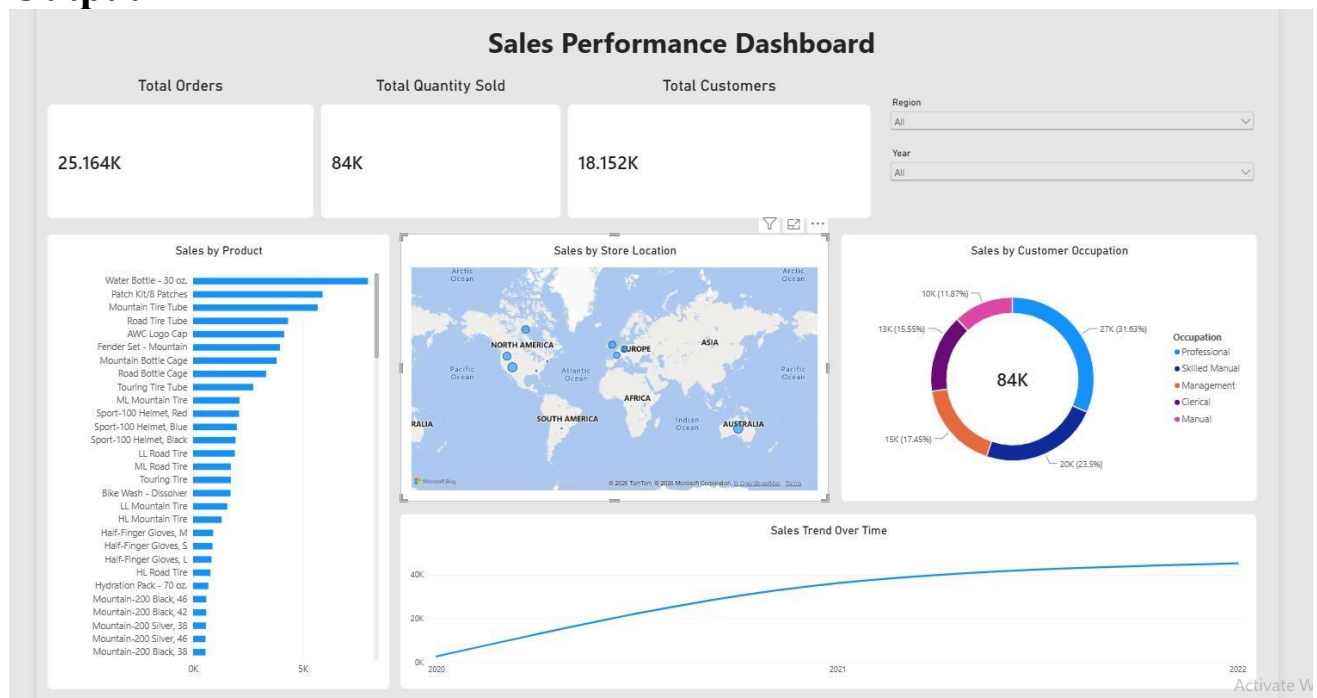


Figure 9: AdventureWorks Sales Performance Dashboard — Report View

The final Power BI dashboard titled "AdventureWorks Sales Performance Dashboard" successfully visualizes the retail company's business data. The dashboard provides:

- **Total Orders:** 25.164K orders recorded across 2020-2022
- **Total Quantity Sold:** 84K units sold across all product categories
- **Total Customers:** 18.149K unique customers served
- **Product Performance:** Water Bottle 30oz, Patch Kit, and Mountain Tire Tube are the top selling products
- **Customer Segments:** Professional customers contribute the highest sales at 31.63%
- **Sales Growth:** Consistent upward trend in sales from 2020 to 2022
- **Geographic Reach:** Sales recorded across North America, Europe, and Australia
- **Interactive Filters:** Region and Year slicers enable dynamic data exploration

Conclusion

This experiment demonstrated how data from **CSV files** and a **MySQL database** can be **integrated and analyzed using Power BI Desktop**. The AdventureWorks sales data was imported,

transformed, and organized into a structured data model by connecting the Sales table with relevant lookup tables. A star schema was created to support effective data analysis and filtering. The final dashboard presents sales information across **products, customers, regions, and different time periods**, providing a clear and interactive way to understand business performance and support informed decision-making.

Business Insights

The following key business insights were derived from the AdventureWorks Sales Performance Dashboard:

- **Product Demand:** A small group of products contributes significantly to the overall quantity sold, showing which items have stronger customer demand.
- **Customer Profile:** The Professional category forms the largest portion of the customer base by sales, suggesting that business-focused customers are an important source of revenue.
- **Performance Over Time:** The sales pattern improves across the three-year period, indicating that the business maintained positive momentum.
- **Regional Contribution:** North America contributes the largest share of the business activity, while the other regions provide additional growth opportunities.
- **Overall Sales Activity:** The large number of orders and units purchased indicates consistent customer purchasing activity throughout the period.
- **Dashboard Filtering:** Users can select a particular year or region to examine changes in performance instead of relying only on the overall totals.